

AGENTS

Base agent rules: `constitution/AGENTS.md` — READ IT FIRST. The base file is authoritative for any topic not covered here. Project-specific rules below extend them; they never weaken them.

Locate the constitution submodule from any nested depth with its `constitution/find_constitution.sh` helper. Canonical reference: <https://github.com/HelixDevelopment/HelixConstitution>

This file contains project-specific information intended for AI coding agents. The reader is expected to know nothing about the project beforehand.

Project Overview

This is a Go-based ebook translation system that translates books between multiple languages using various LLM providers. It supports multiple input/output formats (FB2, EPUB, TXT, HTML, PDF, DOCX) and provides both CLI tools and API servers for translation workflows.

Key capabilities:

- Multi-format ebook parsing and generation
 - Integration with multiple API LLM providers (OpenAI, Anthropic, DeepSeek, Zhipu, Qwen, Gemini, and 20+ more via the LLMsVerifier bridge). **As of bridge phase-2 (R-2..R-4) the local-runtime providers (Ollama, llama.cpp) and the SSH-local translation path were removed** — the default translator path sources the strongest verified API model via the LLMsVerifier bridge (`pkg/bridge`); selecting a removed local provider explicitly now returns an honest error, never a silent local fallback (§11.4.69).
 - Distributed translation (kept) via the distributed/API worker coordinator
 - Real-time WebSocket monitoring dashboard
 - REST API and gRPC interfaces
 - Translation caching, quality verification, and multi-pass processing
 - Preparation phase analysis for improved translation quality
 - Markdown workflow (EPUB to Markdown and back)
-

Technology Stack

- **Language:** Go 1.25.2 (per `go.mod`); Dockerfile uses `golang:1.24-alpine` as builder

- **Module:** `digital.vasic.translator`
 - **Version:** Makefile declares `3.0.0`; `VERSION` file reads `2.3.0`
 - **Web Framework:** Gin (github.com/gin-gonic/gin)
 - **WebSocket:** Gorilla WebSocket (github.com/gorilla/websocket)
 - **gRPC:** google.golang.org/grpc with Protocol Buffers
 - **Authentication:** JWT (github.com/golang-jwt/jwt/v5)
 - **Databases:** PostgreSQL (github.com/lib/pq), SQLite (github.com/mattn/go-sqlite3), Redis (github.com/redis/go-redis/v9)
 - **Document Processing:** `unidoc/unioffice`, `unidoc/unipdf/v3`
 - **CLI Framework:** Cobra (github.com/spf13/cobra)
 - **Testing:** testify (github.com/stretchr/testify)
 - **Transport:** QUIC support (github.com/quic-go/quic-go)
 - **TLS:** golang.org/x/crypto
-

Build and Test Commands

Building

```
# Build all primary binaries (gRPC server, API server, unified
# CLI)
make build

# Build specific components
make build-grpc      # ./build/grpc-server
make build-api       # ./build/api-server
make build-cli       # ./build/unified-translator

# Build for all platforms (Linux AMD64/ARM64, macOS AMD64/ARM64,
# Windows AMD64)
make build-all

# Clean build artifacts
make clean
```

Additional entry points that can be built individually:

```
go build -o build/monitor-server ./cmd/monitor-server
go build -o build/translator ./cmd/translator
go build -o build/server ./cmd/server
go build -o build/markdown-translator ./cmd/markdown-translator
go build -o build/preparation-translator ./cmd/preparation-
    translator
go build -o build/ebook-translator ./cmd/ebook-translator
go build -o build/translate-ssh ./cmd/translate-ssh
go build -o build/deployment ./cmd/deployment
```

Testing

```
# Run all tests (excludes build-tagged tests by default)
make test
```

```
# or
go test ./... -v

# Run tests with coverage (generates coverage.html)
make test-coverage

# Run a specific package
go test -v ./pkg/translator
go test -v ./pkg/events
go test -v ./pkg/websocket

# Run a specific test function
go test -v -run TestFunctionName ./pkg/package

# Quick development cycle (format + vet + test)
make quick-test

# Pre-commit checks
make pre-commit
```

Running Tagged Tests

Tests outside the default cycle are gated by build tags:

```
# Integration tests
go test -tags=integration -v ./test/integration/...

# End-to-end tests
go test -tags=e2e -v ./test/e2e/...

# Performance benchmarks
go test -tags=performance -v ./test/performance/...

# Stress tests
go test -tags=stress -v ./test/stress/...

# Security tests
go test -tags=security -v ./test/security/...
```

Code Quality

```
# Format code
make fmt

# Vet code
make vet

# Lint (requires golangci-lint installation)
golangci-lint run
# Configuration is in .golangci.yml
```

Running Services (Development)

```
# Start development environment (gRPC on :50051 + API on :8080 in
  debug mode)
make dev

# Run full system (builds and runs both servers)
make run-system

# Individual servers
make run-grpc    # gRPC server on port 50051
make run-api     # API server on port 8080

# WebSocket monitoring server
go run ./cmd/monitor-server
# Dashboard available at http://localhost:8090/monitor
```

Docker

```
# Build Docker image
make docker-build

# Run Docker container
make docker-run

# Full stack with PostgreSQL and Redis
docker-compose up -d
```

Code Organization

Directory Structure

cmd/	# Command-line entry points (14 binaries)
api-server/	# REST API server (port 8080)
challenge-runner/	# Runs project challenges/validations
cli/	# Legacy CLI tool
deployment/	# Deployment management tool
ebook-translator/	# Specialized SSH ebook translator
grpc-server/	# gRPC translation service (port 50051)
markdown-translator/	# Markdown workflow tool
monitor-server/	# WebSocket monitoring server (port 8090)
preparation-translator/	# Pre-translation analysis tool
server/	# Main REST API server (older entry point, TLS/HTTP3)
ssh-translation/	# Complete SSH-based translation system
translate-ssh/	# SSH worker standalone binary
translator/	# Legacy translator CLI
unified-translator/	# Primary unified CLI with full provider support

pkg/	# Public packages
api/	# REST API handlers, middleware, routing
batch/	# Batch processing logic
coordination/	# Multi-LLM coordination
deployment/	# Docker and SSH deployment utilities
distributed/	# Distributed processing over SSH workers
ebook/	# Universal ebook parsing (FB2, EPUB, TXT,
HTML, PDF, DOCX)	
events/	# Event bus system for real-time updates
fb2/	# FB2 format-specific handling
format/	# Format detection and validation
grpc/	# gRPC service implementation and proto
definitions	
hardware/	# Hardware detection for optimization
hash/	# Codebase hash utilities
language/	# Language detection utilities
logger/	# Structured logging utilities
markdown/	# EPUB to Markdown conversion workflow
models/	# Data models and registry
preparation/	# Pre-translation preparation phase
progress/	# Progress tracking
report/	# Report generation
script/	# Script conversion (Cyrillic/Latin)
security/	# JWT authentication, rate limiting
sshworker/	# SSH worker management
storage/	# Database abstraction (PostgreSQL, Redis,
SQLite)	
translator/	# Translation engine core
llm/	# LLM provider implementations
verification/	# Translation quality verification
version/	# Version information
websocket/	# WebSocket hub and connection management
internal/	# Internal packages
cache/	# In-memory caching layer
config/	# Configuration loading and validation
scripts/	# Production and deployment shell scripts
working/	# Working configuration files and runtime
data	
test/	# Test suites organized by scope
distributed/	# Distributed system tests
e2e/	# End-to-end tests
fixtures/	# Test data (ebooks, configs, translations)
integration/	# Cross-package integration tests
mocks/	# testify/mock implementations
performance/	# Benchmarks and performance tests
security/	# Security-focused tests
stress/	# Load and stress tests
translator/	# Translator-specific tests
unit/	# Unit tests for individual components
utils/	# Test helper utilities and infrastructure
api/	# API documentation

examples/	# API usage examples
openapi/	# OpenAPI 3.0 specification
web/	# Web interface assets
templates/	# HTML templates (dashboard.html)
scripts/	# Utility and demonstration scripts
host-power-management/	# CONST-033 enforcement scripts
demo-all.sh	
ebook_translation_workflow.sh	
python_translation.sh	
run_monitoring_demo.sh	
build/	# Build artifacts directory
docs/	# Project documentation
Documentation/	# Additional documentation and guides
challenges/	# Submodule: challenge framework
containers/	# Submodule: container runtime extensions

Code Style Guidelines

Go Conventions

- **Naming:** PascalCase for exported identifiers, camelCase for unexported.
- **Imports order:** Standard library → third-party → local packages (digital.vasic.translator/...). Keep alphabetical within each group.
- **Types:** Both interface{} and any appear in the codebase; interface{} is still prevalent in many packages (e.g., map[string]interface{} for config options and event data).
- **Error handling:** Always handle errors explicitly. Wrap errors with context: `fmt.Errorf("operation failed: %w", err)`.
- **Comments:** Document all exported functions, types, and packages. Avoid commenting the obvious.
- **Line length:** Keep lines under 140 characters (per `.golangci.yml`).
- **Function length:** Aim for under 150 lines or 120 statements (funlen setting). Tests and cmd/ entry points have relaxed rules.
- **Cyclomatic complexity:** gocyclo minimum is 35 in config; gocognit minimum is 20.

Security Rules

- **Never hardcode API keys or secrets.** Use environment variables or secure config files.
- **Never commit files containing secrets.** `.gitignore` already excludes `config_with_keys.json`, `.env`, `secrets.*`, etc.
- Input validation is required on all API endpoints.
- Use parameterized queries or ORM patterns for database access.

Linting Configuration

The `.golangci.yml` file enables a strict set of linters including:

- `govet`, `staticcheck`, `gosimple`, `ineffassign` for correctness
- `gocyclo`, `funlen`, `gocognit`, `nestif` for complexity
- `gosec` for security issues
- `goimports` with local prefix `digital.vasic.translator`
- `misspell` with US English locale

Test files and `cmd/` entry points have relaxed rules for `funlen`, `gochecknoinits`, `gosec`, `errcheck`, etc.

Testing Strategy

Test Organization

Tests are organized by scope rather than always co-located with source files:

- `test/unit/` - Unit tests for individual packages and components
- `test/integration/` - Cross-package integration tests (build tag: `integration`)
- `test/e2e/` - End-to-end tests with realistic scenarios (build tag: `e2e`)
- `test/security/` - Authentication, authorization, and security tests (build tag: `security`)
- `test/performance/` - Benchmarks and performance tests (build tag: `performance`)
- `test/stress/` - Load and stress tests (build tag: `stress`)
- `test/distributed/` - Distributed system and SSH worker tests
- `test/mocks/` - Shared mock implementations using `testify/mock`
- `test/fixtures/` - Sample ebooks, configs, and translation data
- `test/utils/` - Shared test helpers and infrastructure

Many packages also have `*_test.go` files alongside source code, particularly in `pkg/api/`, `pkg/translator/`, `pkg/distributed/`, `cmd/`, and `internal/`.

Testing Patterns

- **Table-driven tests:** Preferred style. Define a slice of anonymous structs with inputs and expected outputs, then iterate with `t.Run()`.
- **Sub-tests:** Use `t.Run("descriptive_name", func(t *testing.T) { ... })` for clarity.
- **Mocking:** Use `testify/mock` for interface mocking. Mocks are defined in `test/mocks/providers.go`.
- **Build tags:** Use `//go:build integration`, `//go:build e2e`, `//go:build performance`, `//go:build stress`, `//go:build security` for tests that should not run in the default quick-test cycle.
- **HTTP testing:** Use `httptest` package and `gin.SetMode(gin.TestMode)` for API handler tests.
- **SSH testing:** `test/utils/ssh_test_server.go` provides a full in-process SSH server with generated RSA keys for distributed tests.

- **Short mode:** Many tests check `testing.Short()` and skip when `-short` is used.

Test Utilities

- `test/utils/helpers.go` provides helpers for creating temporary test files, test configs, and minimal valid ebooks (EPUB, FB2, TXT, HTML).
- `test/utils/test_infrastructure.go` provides `TestHTTPServer` and `TestWebSocketServer` for integration tests.
- `test/utils/ports.go` manages dynamic port allocation starting from 30000.
- `test/mocks/providers.go` provides `MockTranslator`, `MockLLMProvider`, `MockDatabase`, `MockSecurityProvider`, `MockStorage`, and `MockProgressReporter`.

Coverage

- **Tool:** Go's built-in coverage (`go test -cover`)
 - **Makefile target:** `make test-coverage`
 - **Artifacts:** `coverage.out` and `coverage.html` are generated at project root
 - **Current coverage:** Approximately 43.6% overall
-

Architecture and Key Patterns

Event-Driven Architecture

The system uses a central event bus (`pkg/events/events.go`) for component communication and real-time progress tracking:

- **Event types:** `EventTranslationStarted`, `EventTranslationProgress`, `EventTranslationCompleted`, `EventTranslationError`, `EventConversionStarted`, `EventConversionProgress`, `EventConversionCompleted`, `EventConversionError`.
- **Subscription:** `Subscribe(eventType, handler)` for type-specific; `SubscribeAll(handler)` for global listeners.
- **WebSocket Integration:** `pkg/websocket/hub.go` subscribes to all events and broadcasts to connected dashboard clients.
- **Session tracking:** Every translation operation has a unique `SessionID`. Events are tagged with session IDs so the dashboard can filter per-client.
- **Concurrency:** Every event handler is invoked in its own goroutine with panic recovery.

Translation Pipeline

1. **Format detection** (`pkg/format/detector.go`) identifies the input file type by magic bytes, extension, and content analysis.
2. **Parsing** (`pkg/ebook/parser.go` and format-specific parsers) extracts content into a Book model.
3. **Preparation** (optional, `pkg/preparation/`) analyzes content for characters, terminology, culture, and chapter structure using multi-pass LLM analysis.

4. **Translation** (pkg/translator/translator.go and pkg/translator/llm/) sends segments to LLM providers with smart retry and automatic chunking on token limit errors.
5. **Verification** (pkg/verification/) runs quality checks (untranslated block detection, HTML artifact detection, completeness) and supports multi-LLM consensus polishing.
6. **Output generation** writes translated content back to the target format (EPUB writer, Markdown converter).

LLM Provider Pattern

All LLM providers implement the LLMClient interface in pkg/translator/llm/llm.go:

```
type LLMClient interface {  
    Translate(ctx context.Context, text string, prompt string)  
        (string, error)  
    GetProviderName() string  
}
```

Supported providers:

- API-based: OpenAI, Anthropic, Zhipu, DeepSeek, Qwen, Gemini (+ 20+ more reachable via the LLMsVerifier bridge / verified discovery)
- Test: Mock provider
- **Removed (bridge phase-2 R-2..R-4):** Ollama (local HTTP) and LlamaCpp (local binary) self-hosted runtimes, and the SSH-local remote-worker path, are no longer supported. Explicit selection (`-provider ollama|llamacpp|ssh`) returns an honest error — there is NEVER a silent local fallback (§11.4.69). The KEPT distributed/API worker coordination path remains.

Default path = the LLMsVerifier bridge. The default translator provisioning across the binaries (unified-translator, cmd/cli, cmd/server / pkg/api, grpc-server, markdown-translator, preparation-translator) sources the strongest verified API model + fallback chain via pkg/bridge (bridge.Open → BestTranslator/BestClient); the mock seam is the only deliberate bypass. With no provider API key the bridge hard-errors honestly. The pre-build gate **CM-NO-LOCAL-RUNTIME** (§11.4.69) mechanically asserts no local-runtime client is constructed on the default path, the default arm references the bridge, and pkg/bridge retains its no-fail-open prohibition (paired §1.1 mutation: scripts/testing/meta_test_no_local_runtime.sh).

Provider selection uses a factory pattern (NewLLMTranslatorWithConfig()) that validates provider/model and creates the appropriate client (used by the mock seam + verified factory). Valid models are registered in ValidModels map.

Import cycle avoidance: pkg/translator/llm/ uses a type alias (type TranslationConfig = translator.TranslationConfig) to avoid direct import cycles with pkg/translator/.

Configuration System

- `internal/config/config.go` defines the canonical config struct with nested sections: Server, Security, Translation, Preparation, Distributed, Logging.
- Config is loaded from JSON files (e.g., `config.json`, files in `internal/working/`).
- Environment variables override file values, especially for secrets and API keys (`OPENAI_API_KEY`, `ANTHROPIC_API_KEY`, `ZHIPU_API_KEY`, `DEEPSEEK_API_KEY`, `JWT_SECRET`).
- `Config.Validate()` checks ports, TLS requirements, JWT secret length (minimum 16 characters), and distributed worker configs.

Distributed Processing

- `pkg/distributed/coordinator.go` manages remote LLM instances across SSH workers with round-robin selection and fallback strategies.
- `pkg/distributed/manager.go` is the top-level orchestrator composing `SSHPool`, `PairingManager`, `VersionManager`, `FallbackManager`, and `DistributedCoordinator`.
- `pkg/sshworker/worker.go` handles SSH connection management, file transfer, remote command execution, and codebase version synchronization.
- `pkg/distributed/version_manager.go` compares local vs. remote codebase versions, supports update/rollback flows, and emits drift alerts.
- `pkg/distributed/fallback.go` tracks per-component failure rates with exponential backoff and degraded mode fallback (remote → local → reduced quality).
- Workers must be reachable from the coordinator. Version synchronization is required.

Markdown Workflow

- `pkg/markdown/epub_to_markdown.go` converts EPUB to Markdown preserving metadata, cover images, and chapter structure.
- `pkg/markdown/markdown_to_epub.go` converts translated Markdown back to EPUB.
- `pkg/markdown/translator.go` translates Markdown line-by-line while preserving syntax (headers, lists, code blocks, links, inline formatting).
- `pkg/markdown/simple_workflow.go` orchestrates the full Ebook → Markdown → Translate → Ebook pipeline.

gRPC and Protobuf

- Protobuf definition lives at `pkg/grpc/translator.proto`.
 - Generated code is consumed from `pkg/grpc/proto`.
 - `pkg/grpc/server.go` implements `TranslationService` with session management, streaming progress, provider registry, and cleanup routines.
-

Security Considerations

- **TLS required for production:** The main server entry point uses HTTPS/HTTP3. TLS certificates are expected in the `certs/` directory (`server.crt`, `server.key`). Self-signed cert generation is available via `--generate-certs` flag.
- **JWT authentication:** Configurable via `security.enable_auth` in config. Secret must be at least 16 characters. Timing attack mitigation adds an artificial delay for invalid tokens.
- **Rate limiting:** Configurable requests-per-second and burst limits via `security.rate_limit_rps` and `security.rate_limit_burst`. Per-key limiters with background cleanup every 10 minutes.
- **API key management:** API keys must be provided via environment variables (e.g., `OPENAI_API_KEY`, `ANTHROPIC_API_KEY`). Never commit them.
- **CORS:** Configurable allowed origins via `security.cors_origins`.
- **Input validation:** All API endpoints validate request bodies using Gin binding tags (`binding:"required"`, `binding:"email"`, etc.).
- **SSH security:** Key-based authentication is preferred for SSH workers. Password auth is supported but discouraged for production.
- **Transport:** HTTP/3 uses TLS 1.3 with h3 ALPN; HTTP/2 fallback uses TLS 1.2+.

Sensitive Files (Already in .gitignore)

- `config_with_keys.json`, `api_keys.json`
 - `.env`, `.env.*`, `secrets.*`, `keys.*`
 - `qwen_credentials.json`, `oauth_creds.json`
 - `.translator/` directory
-

Deployment Processes

Docker Deployment

A multi-stage Dockerfile is provided:

- **Builder stage:** `golang:1.24-alpine` with `git` and `make`
- **Production stage:** `alpine:latest` with `ca-certificates` and `tzdata`
- Runs as non-root user (`appuser`, UID 1000)
- Exposes port 8443 (TCP and UDP for HTTP/3)
- Health check endpoint: `https://localhost:8443/health`
- Default command: `/app/translator-server -config /app/config/config.json`
- Volumes exposed for `/app/certs` and `/app/config`

Note: The Dockerfile references `make build-server`, which does not exist in the root Makefile. The Makefile defines `build`, `build-grpc`, `build-api`, and `build-cli` instead.

Docker Compose

`docker-compose.yml` provides a full production stack:

- PostgreSQL 16 (with health checks and optional init script)
- Redis 7 (with password, 512MB maxmemory, allkeys-lru policy)
- Translator API server (depends on healthy postgres & redis)
- Optional admin tools (Adminer, Redis Commander) via `--profile admin`

`docker-compose.distributed.yml` provides a distributed worker stack with main server, worker instances, and Ollama services.

Manual Deployment

```
# Build binaries
```

```
make build
```

```
# Start gRPC server
```

```
./build/grpc-server
```

```
# Start API server
```

```
./build/api-server
```

```
# Start monitoring server
```

```
./build/monitor-server
```

Deployment Scripts

`internal/scripts/` contains extensive operational scripts:

- `deploy-system.sh` - Automated deployment using `./build/deployment-cli`
 - `deploy_ssh_worker.sh` / `deploy_worker.sh` - SSH worker deployment
 - `check-health.sh` / `check_worker.sh` - Health checks
 - `monitor-production.sh` - Full production monitoring with disk/CPU/memory/API/DB/Redis checks and alerting
 - `start.sh` / `stop.sh` / `restart.sh` / `logs.sh` / `exec.sh` - Docker lifecycle management
 - `demo_production_system.sh` / `test_production_system.sh` - Production verification suites
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Development Conventions

Adding a New LLM Provider

1. Create a new file under `pkg/translator/llm/<provider>.go`.
2. Implement the `LLMClient` interface with HTTP POST JSON requests (or local execution for self-hosted).
3. Add valid models to `ValidModels` map in `pkg/translator/llm/llm.go`.

4. Add provider-specific configuration to `internal/config/config.go` if needed.
5. Register the provider in the factory switch in `NewLLMTranslatorWithConfig()`.
6. Add unit tests in `test/unit/` or alongside the package.
7. Update `api/openapi/openapi.yaml` if the provider appears in API enums.

Adding a New File Format

1. Implement the parser interface (`Parse(filename string) (*Book, error)`) in `pkg/ebook/`.
2. Register the format in `pkg/ebook/parser.go` via `RegisterParser()`.
3. Add detection logic in `pkg/format/detector.go`.
4. Add writer support if output generation is required.
5. Add sample files to `test/fixtures/ebooks/`.
6. Add parser tests.

Session and Progress Tracking

- Always generate a unique `SessionID` (e.g., `UUID`) for each translation operation.
- Emit progress events via the event bus so `WebSocket` clients receive updates.
- The monitoring dashboard (`web/templates/dashboard.html`) displays sessions, progress bars, event logs, and worker status.

Goroutine and Context Patterns

- `context.Context` is the first parameter in virtually all I/O and long-running operations.
 - Use `context.WithTimeout` for API calls; `context.WithCancel` for per-session cancellation.
 - Always defer `mu.Unlock()` after `mu.Lock()` or `mu.RLock()`.
 - `WebSocket` and event handlers launch background goroutines with panic recovery.
 - Graceful shutdown catches `SIGINT/SIGTERM` and cancels contexts with a 30-second timeout.
-

Critical Non-Negotiable Constraints

- **LLMsVerifier SSOT**: `LLMsVerifier` is the EXCLUSIVE source of truth for all LLM models. No hardcoded model lists. No unverified models. No bypass.
- **HELIXQA ONLY**: All automated UI/UX testing must use HelixQA. No custom browser scripts.
- **ANTI-BLUFF MANDATE**: Every test must verify real user-visible behavior. See `CONSTITUTION.md` `CONST-035`.
- **FULL-QA MASTER CYCLE**: Every change requires the full Article VII cycle.
- **4-ARTEFACT FIX**: Every defect fix must produce: unit test + integration test + bank entry + challenge.

- **EVIDENCE OR BUST:** Passing tests without captured evidence = invalid. Screenshot/video/log required.

HelixQA Commands

```
# Build HelixQA binary
cd HelixQA && make build

# Standard QA
helixqa run --banks tests/banks/ --platform all --speed normal

# List available tests
helixqa list --banks tests/banks/ --platform web --json

# Autonomous QA session
helixqa autonomous --project . --platforms web,api,cli \
  --timeout 2h --output qa-results/ --verbose

# Generate report
helixqa report --input qa-results/ --format html --output qa-
report.html
```

QA Bank Format (YAML)

```
version: "1.0"
name: "HelixTranslate Full QA - API"
test_cases:
  - id: HTQ-API-001
    name: "Health check returns healthy status"
    category: functional
    priority: critical
    platforms: [api]
    steps:
      - name: "Send GET /health"
        action: "GET https://localhost:8443/health"
        expected: "HTTP 200 with {\"status\":\"healthy\"}"
    tags: [health, smoke]
    expected_result: "Health endpoint returns valid JSON with
      healthy status"
```

HelixQA Ecosystem (Sibling Modules)

The project depends on several sibling Go modules managed as Git submodules:

./HelixQA	→ digital.vasic.helixqa	(QA automation engine)
./DocProcessor	→ digital.vasic.docprocessor	(Document processing)
./LLMOrchestrator	→ digital.vasic.llmorchestrator	(Agent)

orchestration)		
./LLMProvider	→ digital.vasic.llmprovider	(Provider facade
+ circuit breaker)		
./VisionEngine	→ digital.vasic.visionengine	(Vision + OpenCV
analysis)		
./LLMsVerifier	→ digital.vasic.llmsverifier	(Model
verification SSOT)		
./Challenges	→ digital.vasic.challenges	(Challenge
framework)		
./Containers	→ digital.vasic.containers	(Container
orchestration)		
./Models	→ digital.vasic.models	(Shared data
types)		
./Security	→ digital.vasic.security	(Security
utilities / SSRF)		

All submodules are wired via replace directives in go.mod. If a submodule fails to clone (e.g. LLMsVerifier repository unavailable), create a local placeholder with a matching go.mod module name and populate the packages that HelixTranslate imports.

LLMsVerifier Integration Architecture

```

[HelixTranslate Application]
|
| 1. Config Load
v
[internal/verifier/]
|
| 2. Verify All Providers
v
[digital.vasic.llmsverifier/pkg/api]
|---> types.go (canonical Model, PricingInfo – CONST-034 SSOT)
|---> pkg/helixqa (VisionModelRegistry for HelixQA ranking)
|
| 3. Return Verified + Scored Models
v
[internal/services/llmsverifier_score_adapter.go]
|
| 4. Score Normalization (0-100 -> 0-10)
v
[pkg/translator/llm/llm.go]
|
| 5. Translation via Verified Providers (26 providers)
v
[LLM Provider APIs]

```

Canonical type location: llms_verifier/pkg/api/types.go defines Model and PricingInfo. HelixTranslate aliases them in internal/verifier/client.go (type Model = api.Model).

Essential Files Quick Reference

File	Purpose
go.mod / go.sum	Module definition and dependency checksums
Makefile	Build, test, and development commands
config.json	Default application configuration
Dockerfile	Multi-stage container build
docker-compose.yml	Full stack with PostgreSQL and Redis
.golangci.yml	Strict linting configuration
internal/config/config.go	Canonical configuration structs and loading
pkg/events/events.go	Central pub/sub event system
pkg/websocket/hub.go	WebSocket hub for real-time monitoring
pkg/translator/llm/llm.go	LLM provider interface and factory (26 providers)
llms_verifier/pkg/api/types.go	Canonical model verification types (CONST-034 SSOT)
api/openapi/openapi.yaml	OpenAPI 3.0 specification
VERSION	Current application version (2.3.0)

Common Pitfalls

- **FB2 XML:** Always register the FB2 namespace before parsing. Use UTF-8 encoding. Preserve XML hierarchy during translation.
- **LLM Rate Limits:** Each provider has different limits. Implement backoff and respect max_concurrent settings.
- **SSH Workers:** Ensure key-based auth is configured. Workers must run the same binary version as the coordinator.
- **Large Files:** Use streaming for large ebook translations to avoid excessive memory usage.
- **TLS Certificates:** The API server requires valid certificates in certs/ for HTTPS/HTTP3. Generate or provide them before starting the server.
- **Import Cycles:** The pkg/translator/llm/ package uses a type alias to avoid cycles with pkg/translator/. Be careful when adding new cross-package dependencies.
- **Version Discrepancies:** The Makefile declares 3.0.0, the VERSION file reads 2.3.0, and go.mod specifies Go 1.25.2 while the Dockerfile uses golang:1.24-alpine. Treat these as the current state of the project.
- **No Root-Level CI/CD:** There are no .github/workflows/ at the project root. Validation relies on Makefile targets and shell scripts.

Host Power Management — Hard Ban (CONST-033)

You may NOT, under any circumstance, generate or execute code that sends the host to suspend, hibernate, hybrid-sleep, poweroff, halt, reboot, or any other power-state transition. This rule applies to:

- Every shell command you run via the Bash tool.
- Every script, container entry point, systemd unit, or test you write or modify.
- Every CLI suggestion, snippet, or example you emit.

Forbidden invocations (non-exhaustive — see CONST-033 in CONSTITUTION.md for the full list):

- `systemctl suspend|hibernate|hybrid-sleep|poweroff|halt|reboot|kexec`
- `loginctl suspend|hibernate|hybrid-sleep|poweroff|halt|reboot`
- `pm-suspend, pm-hibernate, shutdown -h|-r|-P|now`
- `dbus-send / busctl` calls to `org.freedesktop.login1.Manager.Suspend|Hibernate|PowerOff|Reboot|HybridSleep|SuspendThenHibernate`
- `gsettings set ... sleep-inactive-{ac,battery}-type` to anything but `'nothing'` or `'blank'`

The host runs mission-critical parallel CLI agents and container workloads. Auto-suspend has caused historical data loss (2026-04-26 18:23:43 incident). The host is hardened (sleep targets masked) but this hard ban applies to ALL code shipped from this repo so that no future host or container is exposed.

Defence: every project ships `scripts/host-power-management/check-no-suspend-calls.sh` (static scanner) and `pkg/challenge_runner/scripts/no_suspend_calls_challenge.sh` (challenge wrapper). Both **MUST** be wired into the project's CI / `run_all_challenges.sh`.

Full background: `docs/HOST_POWER_MANAGEMENT.md` and `CONSTITUTION.md` (CONST-033).

CONST-035 — Anti-Bluff Operative Rule (MANDATORY)

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT** be the case."

The operative rule: Execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product.

- A green test or challenge for a feature that does not actually work is a **BLUFF** and is **FORBIDDEN**.
- Every test must assert concrete user-visible outcomes, not just internal state.
- Every challenge must run real code and verify real behavior; `grep/file-existence` checks are **NOT sufficient**.

- Mutation testing is **MANDATORY**: deliberately break the feature → the test/challenge **MUST then FAIL**.
- The bar for shipping is **NOT "tests pass"** but **"users can use the feature."**
- No false-success results are tolerable.

Mirrored constitution anchors (§11.4.154–§11.4.157)

Condensed consumer-side mirrors of the four universal anchors most recently added to the constitution submodule (HEAD 9f3147e). Canonical text lives in `constitution/Constitution.md`; these mirrors carry the literal 11.4.N tokens the propagation gates require (§11.4.35).

§11.4.154 — Window-scoped capture + fresh-corpus rotation for feature/QA recordings (User mandate, 2026-06-15). Refines §11.4.2/.5/.107/.153 recording discipline with two capture-hygiene invariants. **(A) Window-scoped, NOT whole-screen** — every feature/QA video **MUST** capture **ONLY** the window/surface of the app/service under test (GUI window / CLI-TUI terminal pane / web tab-viewport / device-emulator frame), **NEVER** the whole desktop/monitor or unrelated windows; target by stable identity (window id/title, device serial, browser context, tmux target) per §11.4.111, never a fixed full-screen index; platform genuinely cannot capture below whole-screen ⇒ honest §11.4.3 SKIP + tracked migration item. **(B) Fresh-corpus rotation** — a new recording run for a scope removes the agent's **OWN** prior in-scope stale recordings **FIRST** (§11.4.107 not-stale + §11.4.86 roster-freshness); the agent's own **SAME**-scope recordings **ONLY**, **NEVER** another project's/operator-authored files (uncertain ⇒ surface, don't delete §11.4.122/§9.2); committed docs/qa/<run-id>/evidence (§11.4.83) is the durable record, **NOT** rotated. Classification: universal (§11.4.17). Composes §11.4.2/.5/.10/.83/.86/.107/.111/.122/.128/.137/.153/§9.2/.6. Canonical authority: constitution submodule `Constitution.md` §11.4.154. Non-compliance is a release blocker.

§11.4.155 — Project-name-prefixed feature/QA recording filenames (User mandate, 2026-06-15). Every recorded video the project produces (§11.4.153 real-use, §11.4.154 window-scoped, §11.4.128 always-on device, any raw/curated artefact at the recording path + the committed docs/qa/<run-id>/trail §11.4.83) **MUST** have a filename that **STARTS WITH** the PROJECT-NAME prefix, **ALWAYS**; an unprefixed recording is a §11.4.155 violation (un-greppable + un-attributable multi-project corpus — the §11.4.151 identify-and-grep failure on the recording axis). **Prefix resolution (closed-set, deterministic — §11.4.6, IDENTICAL to §11.4.151):** (1) `HELIX_RELEASE_PREFIX` from `.env` (git-ignored §11.4.30, documented in `.env.example` §11.4.77) else (2) lowercased `snake_case` project-root dir name §11.4.29; canonical form `<PREFIX>---<feature-or-scope>---<run-id>.<ext>`; **MUST** equal the §11.4.151-resolved release-tag prefix. Honest boundary (§11.4.6): the prefix guarantees attribution + greppability, **NOT** content validity (still §11.4.107/.137/.153). Classification: universal (§11.4.17). Composes §11.4.151/.128/.153/.154/.111/.83/.6/.29/.30/.35/.77/.86. Canonical authority: constitution submodule `Constitution.md` §11.4.155. Non-compliance is a release blocker.

§11.4.156 — All CI/CD automation (GitHub Actions / GitLab pipelines / equivalents) MUST be disabled (User mandate, 2026-06-15). Every repository this Constitution governs **MUST** ship with **ALL** server-side CI/CD automation **DISABLED**: no push to any owned upstream may trigger a GitHub Actions run,

GitLab pipeline, or equivalent (Jenkins/CircleCI/Travis/Drone/Woodpecker/Bitbucket/Azure, any on: push/schedule/workflow_dispatch). GENERALISES + makes ABSOLUTE the §11.4.75 Layer-5 posture across ALL governed repos; enforcement migrates to the LOCAL §11.4.75 git-hook ritual + §11.4.40 pre-tag sweep, never a remote runner. ALL hold: **(A)** zero active root-level .github/workflows/*.yaml|yml / .gitlab-ci.yml / equivalent; **(B)** "disabled" = a push triggers ZERO runs — delete OR rename to a non-trigger name (the §11.4.75 .disabled-local-only pattern); a live-on:+if:false workflow is NOT compliant; **(C)** scope = repos we author+push — vendored/third-party nested configs are INERT, OUT of scope (§11.4.29 vendor-exempt); **(D)** no new CI may be added; **(E)** pre-push verify the tracked workflow set is empty for authored repos. Honest boundary (§11.4.6): file-level disabling stops FILE-triggered runs, NOT provider-side server settings — the operator turns those off. Composes §11.4.75/.29/.6/.40/.42/.109/.113/§2.1. Classification: universal (§11.4.17). The parent repo currently has NO active CI workflow files — already §11.4.156-compliant. Canonical authority: constitution submodule Constitution.md §11.4.156. Non-compliance is a release blocker.

§11.4.157 — GEMINI.md maintained in lockstep with CLAUDE.md / AGENTS.md / QWEN.md (User mandate, 2026-06-15). GEMINI.md is a FIRST-CLASS governance context carrier EQUAL to CLAUDE.md/AGENTS.md/QWEN.md, never optional/best-effort. ALL hold: **(A)** five-carrier lockstep — no governance change is complete until GEMINI.md carries it alongside the other three mirrors (added to the §11.4.26 propagation + cross-reference set); **(B)** no silent drift — GEMINI.md lagging the other mirrors' highest rule is a §11.4.157 violation (§11.4.65-class); **(C)** equal status — GEMINI.md restates the SAME literal 11.4.N anchors the propagation gates require (§11.4.35), fleet count INCLUDES GEMINI.md; **(D)** consumer projects' own CLAUDE/AGENTS/QWEN/GEMINI bind too (§11.4.35). Honest boundary (§11.4.6): claiming GEMINI.md "in sync" while a back-fill is incomplete is itself a §11.4.157 violation. Composes §11.4.26/.35/.17/.44/.65/.140/.156. Classification: universal (§11.4.17). This parent repo now maintains CLAUDE.md + AGENTS.md + QWEN.md + GEMINI.md in lockstep (QWEN.md + GEMINI.md created in this change-window). Canonical authority: constitution submodule Constitution.md §11.4.157. Non-compliance is a release blocker.